2025 AMC 10B Problem 18 - Solution

Review the full statement and step-by-step solution for 2025 AMC 10B Problem 18. Great practice for AMC 10, AMC 12, AIME, and other math contests

2025 AMC 10B Problem 18

Problem:

What is the ones digit of the sum

$$\left\lfloor \sqrt[k]{1} \right\rfloor + \left\lfloor \sqrt[k]{2} \right\rfloor + \left\lfloor \sqrt[k]{3} \right\rfloor + \cdots + \left\lfloor \sqrt[k]{2025} \right\rfloor ?$$

(Recall that $\lfloor x \rfloor$ denotes the greatest integer less than or equal to x.)

Answer Choices:

A. 1

B. 2

C. 3

D. 5

E. 8

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Solution:

We want the ones digit of

$$S = \left\lfloor \sqrt{1} \right\rfloor + \left\lfloor \sqrt{2} \right\rfloor + \cdots + \left\lfloor \sqrt{2025} \right\rfloor .$$

For each positive integer m, we have

$$m^2 \leq n < (m+1)^2 \quad \Longrightarrow \quad \lfloor \sqrt{n} \rfloor = m.$$

On this interval, there are $(m+1)^2 - m^2 = 2m+1$ integers n, each contributing m to the sum. The largest perfect square ≤ 2025 is

$$45^2 = 2025.$$

Thus, the sum is

$$S = 45 + \sum_{m=1}^{44} m(2m+1) = 45 + 2\sum_{m=1}^{44} m^2 + \sum_{m=1}^{44} m = 45 + 2\frac{(44)(44+1)(2\cdot 44+1)}{6} + \frac{44\cdot(44+1)}{2}$$

Since we only need the ones digit, we compute everything modulo 10.

$$S \equiv 45 + 2\frac{(44)(44+1)(2\cdot 44+1)}{6} + \frac{44\cdot(44+1)}{2} \mod 10.$$

Therefore,

$$S \equiv 45 + (44)(15)(89) + 22 \cdot 45 \equiv \boxed{\textbf{(D) } 5} \bmod 10.$$